

Algorithm Problem due to grids

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Problem

Dr. Daripa, I have identified a problem/feature of the algorithms in response to nonuniform grids. For the following, I am referring to meshes where the azimuthal mesh is fixed across radii (but not necessarily azimuthally uniform). Tests are also done on the dirichlet problem as my neumann solver for the nonuniform cases still appears to have an error which I am attempting to fix. Although, the following may explain part or all of that error. I ran some tests for an example problem for varying N and varying M on different grid types. The results are in the tables at the end of this report. Here is what I observed regarding the meshes

Grids and Performance

- **Uniform.** When the grid is uniform, the uniform, NUDFT, and NUFFT algorithms are identical in errors and convergence for both varying M and N .
- **Jittered grids.** For jittered grids (uniform mesh with some small random perturbations applied to each angle), the same results hold as in the uniform case. Although for cases I tested with large N , performance degrades very slightly but it remains good.
- **Random grids.** For fully random grids, the angular nodes are drawn randomly from $\text{Uniform}(0, 2\pi)$ and then sorted. In this case the NUDFT performs very randomly for varying N . The NUDFT performs the same but very badly for fixed but varying M . In the NUFFT case, the error slowly but barely improves for varying N , and the performance is the same but poor for varying M .
- **Clustered grids.** For clustered grids, the angular nodes are generated by a power-law mapping that places many points near a chosen angle and relatively few points elsewhere. For varying N , the NUDFT errors for the clustered case grow significantly with increasing N , and are much larger than the uniform and jittered columns. For a fixed N , the NUDFT errors for clustered grids stay nearly constant (but poor) as M increases. For the NUFFT, the same pattern is observed, although performance for varying N degrades much slower.
- **Sine-perturbed grids.** For sine-perturbed grids, the angular nodes start from a uniform mesh and then are smoothly perturbed using a sine function. For varying N , the NUDFT errors in the sine column grow and fluctuate substantially with N . For varying M and fixed N , the NUDFT errors for sine-perturbed grids slowly decay as M increases. For the NUFFT, increasing N decreases error slowly, and fixing N but varying M shows a constant error.

- **Timing.** I also note the FFT and NUDFT are fast, but the NUFFT is especially slow, mostly in cases where the errors were worse (more CG iterations required).

Algorithms

Recall the three azimuthal algorithm choices from the main paper. The three angular algorithms differ only in how they implement the same basic idea (recovering Fourier modes from azimuthal samples):

- **Uniform FFT case.** On a perfectly uniform grid, the angular transform is unitary up to scaling. The forward and inverse transforms are exact and extremely well conditioned, and the FFT computes them directly.
- **NUDFT case.** On a nonuniform grid with fixed angles, the same angular transform becomes a general dense matrix. To go from samples to modes one now has to solve a linear system. The direct NUDFT implementation solves this system exactly (up to roundoff), so it fully exposes any conditioning issues associated with the chosen mesh.
- **NUFFT case.** The NUFFT avoids forming the dense matrix explicitly and instead uses fast Fourier-based routines to apply the forward and adjoint transforms inside an iterative least-squares solver through conjugate gradient method. This produces an approximate solution whose quality depends both on the mesh (through the conditioning of the system) and on the stopping criteria of the iterative method. On “good” meshes the NUFFT reproduces the uniform FFT behavior, while on “bad” meshes it may not be able to fit all modes as accurately as in the uniform case.

Performance Explanation

Here is one possible explanation I have. I do not know for sure that it is correct but it seems plausible. On uniform and jittered grids, the matrices A and $A^{(\ell)}$ are well conditioned for the grids in question. All three algorithms (FFT, NUDFT, NUFFT) effectively compute the same Fourier coefficients, so their errors and convergence closely match. On random, clustered, and sine-perturbed grids, the matrices $A^{(\ell)}$ become increasingly ill-conditioned as N grows (for fixed M). The direct NUDFT inversion fully inverts all singular values, including extremely small ones, so rounding errors and modeling errors are strongly amplified. In the NUFFT case, conjugate gradients solve the normal equations only approximately. With a fixed tolerance and iteration cap, CG captures the well-resolved components of $A^{(\ell)}$ but does not fully invert directions corresponding to the smallest singular values. This explains why, on problematic grids, NUFFT errors may improve slowly with N but do not approach the FFT/NUDFT accuracy achieved on uniform and jittered meshes, and why the NUFFT runtimes are largest precisely in the cases where the errors are worst (more CG iterations required). And for the cases where M grows but N is fixed, no improvement is seen for the random or clustered grids. Perhaps the accuracy is constrained not by M as in the uniform case, but by the accuracy of the Nonuniform azimuthal part of the algorithm.

I recommend that you read the following: <https://finufft.readthedocs.io/en/latest/tutorial/invid2.html#:~:text=conditioning%20of%20the%20inverse%20NUFFT%20problem> The inverse 1D NUFFT tutorial for FINUFFT emphasizes that the conditioning of the inverse problem is set mainly by the distribution of the sample points and by the maximum frequency being fitted, not by the particular solver that is used. In particular, they observe that when the nonuniform

nodes contain a relatively large gap the least-squares system becomes ill-conditioned: the normal-equation matrix has very small singular values, and one can solve the linear system to a small residual while still having noticeably larger errors in the recovered Fourier coefficients and in the reconstructed function near that gap. They point out that this behavior is a feature of the inverse problem itself and is not fixed by changing from a NUFFT-based solver to another method.

If our method is not robust for actual random azimuthal grids that are badly spaced, then I believe we may have an issue. Solving the ill conditioned systems using the NUDFT and NUFFT method both will take much longer times (convergence speed etc...) and limits the accuracy of the method based on how well the NUFFT recovers the data. This is consistent with the prior results weve seen, where the NUFFT method is extremely slow (moreso than the NUDFT) and above where the accuracy of the methods is greatly degraded and capped by the azimuthal portion. Would you be able to look into this either currently or after reading the draft? I'm not sure if there are some approaches we can take to fix this (i've seen references mention regularization or overdetermined systems) or if it is amenable at all. If so, this is something we should dicuss as if not it may impact our ability to write the paper. I can see in either case though that our results would still prove useful in the case where meshes are slightly nonuniform due to sampling inconsistency etc...

Table 1: Table 1: L_∞ errors, $M = 32$, rows N

N	FFT	NUDFT					NUFFT				
	Unif	Unif	Rand	Jitt	Clust	Sine	Unif	Rand	Jitt	Clust	Sine
32	1.9e-05	1.9e-05	3.0e-01	1.9e-05	4.5e-02	1.9e-05	1.9e-05	4.0e-01	1.9e-05	7.2e-03	1.3e-02
64	1.9e-05	1.9e-05	1.8e-05	1.9e-05	4.3e-01	5.8e-01	1.9e-05	3.7e-01	1.9e-05	9.4e-03	5.6e-03
128	1.9e-05	1.9e-05	4.4e+00	1.9e-05	1.9e+00	2.8e-01	1.9e-05	3.3e-01	1.9e-05	1.3e-02	1.5e-03

Table 2: Table 2: L_∞ errors, $N = 32$, rows M

M	FFT	NUDFT					NUFFT				
	Unif	Unif	Rand	Jitt	Clust	Sine	Unif	Rand	Jitt	Clust	Sine
32	1.9e-05	1.9e-05	3.0e-01	1.9e-05	4.5e-02	1.9e-05	1.9e-05	4.0e-01	1.9e-05	7.2e-03	1.3e-02
64	4.5e-06	4.5e-06	3.0e-01	4.6e-06	4.6e-02	4.6e-06	4.5e-06	4.0e-01	4.6e-06	7.2e-03	1.3e-02
128	1.1e-06	1.1e-06	3.0e-01	1.1e-06	4.6e-02	1.1e-06	1.1e-06	4.0e-01	1.1e-06	7.2e-03	1.3e-02

Table 3: Table 3: runtimes (s), $M = 32$, rows N

N	FFT	NUDFT					NUFFT				
	Unif	Unif	Rand	Jitt	Clust	Sine	Unif	Rand	Jitt	Clust	Sine
32	0.01	0.02	0.01	0.02	0.02	0.02	0.10	2.39	2.29	2.70	2.24
64	0.01	0.02	0.01	0.02	0.02	0.03	0.18	4.04	2.01	3.79	2.67
128	0.00	0.04	0.02	0.05	0.07	0.06	0.10	3.56	2.36	4.34	3.64

Table 4: Table 4: runtimes (s), $N = 32$, rows M

M	FFT	NUDFT					NUFFT				
	Unif	Unif	Rand	Jitt	Clust	Sine	Unif	Rand	Jitt	Clust	Sine
32	0.01	0.02	0.02	0.02	0.02	0.03	0.15	3.46	2.13	2.57	2.94
64	0.01	0.03	0.03	0.03	0.03	0.02	0.20	4.79	3.03	3.84	2.83
128	0.02	0.04	0.04	0.04	0.05	0.06	0.26	6.70	4.48	4.69	4.37